Parliamentary Representation, Earmarked Transfers, and Income Inequality in Brazilian States: A Multi-Strategy Causal Analysis

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Abstract

We investigate whether over-representation in the Brazilian Chamber of Deputies — measured by the ratio of state seats to population — increases the volume of earmarked parliamentary amendments (emendas parlamentares) received by states and, ultimately, reduces state-level income inequality (Gini coefficient). Exploiting the constitutionally fixed seat floors (8 seats) and ceilings (70 seats) as an instrumental variable, and applying five distinct causal identification strategies — causal mediation analysis (Imai, Keele, and Tingley, 2010), distributed-lag impulse response functions, Difference-in-Differences around Constitutional Amendment 86/2015, Regression Discontinuity Design at the 8-seat threshold, and Double Machine Learning — we find no robust evidence that parliamentary amendments reduce inequality at the state level. The only statistically significant link detected is between public expenditure on education and the Gini index ($\beta = -0.025$, $p = 0.026$), consistent with a redistribution channel operating through fiscal policy rather than through earmarked transfers per se. Our mediation decomposition reveals a partial-suppression pattern: the indirect effect via amendments (NIE = −0.0031) partially offsets a countervailing positive direct effect (NDE = +0.0058), leaving a statistically indistinguishable-from-zero total effect (ATE = +0.0027, $p = 0.151$). These findings suggest that the scale mismatch between municipal-targeted amendments and state-level inequality measurement may obscure genuine distributional effects that could surface at finer geographic resolution.

Keywords: *parliamentary representation · earmarked transfers · income inequality · causal mediation · instrumental variables · Brazilian fiscal federalism · emendas parlamentares · Double Machine Learning*

JEL Codes: *H72 · H77 · D72 · C26 · N46*

1. Introduction

Brazil's Chamber of Deputies allocates budgetary resources to states through parliamentary amendments — known as emendas parlamentares — which earmark federal funds for projects proposed by individual legislators. Since the Constitutional Amendment 86 of 2015 made a portion of these transfers mandatory (the so-called emendas impositivas), the system has grown substantially: in 2023 alone, total earmarked amendments exceeded R$ 50 billion. This mechanism has attracted considerable policy attention as a potential vehicle for regional redistribution, given that smaller and poorer states are constitutionally over-represented in the Chamber.

The core empirical question is whether this institutional feature — states receiving more deputies per capita — translates into lower income inequality through the mechanism of higher per-capita amendments. While the causal chain has intuitive appeal (more deputies → more earmarked funds → targeted public spending → lower Gini), several confounding forces complicate identification. States with higher per-capita representation tend to be smaller, poorer, and concentrated in the North and Northeast — regions with historically high inequality. Naïve OLS therefore mixes distributional selection with any genuine redistributive effect.

While regional redistribution is not a stated objective of the parliamentary amendment system, an implicit redistributive expectation underlies much of the policy debate: constitutionally over-represented states tend to be smaller and poorer, and it is frequently assumed that their disproportionate legislative weight should translate into fiscal advantage and, ultimately, lower inequality relative to wealthier, under-represented states such as São Paulo. This paper tests that implicit premise empirically.

We address this identification problem through a multi-pronged strategy. Our instrument exploits the constitutional rule that each state receives between 8 and 70 seats regardless of population, creating exogenous variation in per-capita representation. We complement the IV approach with: (i) distributed-lag models that allow for multi-year effect accumulation; (ii) an event study around Constitutional Amendment 86/2015; (iii) an RDD exploiting the 8-seat floor; and (iv) Double Machine Learning (Chernozhukov et al., 2018) to purge confounders non-parametrically. A secondary fiscal mediation chain (amendments → sectoral spending → Gini) is estimated using SICONFI administrative data.

Our main contribution is to provide the first systematic, multi-method causal analysis linking parliamentary representation to inequality in Brazilian states, embedding it within a formal causal graph (DAG) and a mediation

decomposition framework. We also provide, to our knowledge, the first test of the conditional IV validity in this setting via historical development proxies.

2. Related Literature

2.1 Malapportionment and fiscal transfers

The closest international benchmark is Galiani, Galetovic, and Schargrodsky (Economics & Politics, 2016), which analyzes the Brazilian case using IV methods and finds that over-represented states receive disproportionate federal resources, but that this advantage does not translate into higher development — if anything, it may worsen resource allocation. Our paper extends this finding to the inequality dimension and applies a richer identification toolkit.

Beramendi, Rogers, and Díaz-Cayeros (2017) argue that the joint presence of regional inequality and malapportionment distinguishes Latin American federations from other federal systems — less populous (not necessarily poorer) states are the main beneficiaries of skewed transfers. This is the precise mechanism our partial-suppression result captures: over-representation shifts resources to smaller states but through a channel (discretionary earmarks) that does not systematically reduce inequality.

Dunn (2022, Publius) documents that malapportioned legislatures produce a rural-conservative bias that obstructs centralised redistributive reforms. Consistent with this, our direct-effect estimate (NDE = +0.0058) suggests that over-representation marginally raises the state Gini through non-amendment channels, possibly by constraining formula-based equalisation transfers.

2.2 Electoral effects of parliamentary amendments

A large Brazilian literature examines whether parliamentary amendments translate into electoral returns. Ames (1995) and Pereira and Rennó (2007) find positive electoral effects; Samuels (2002) finds null effects; Mesquita (2008) finds negative effects in some contexts. Baião and Couto (2017) reconcile this heterogeneity: only amendments executed as transfers to municipalities generate votes, and only when the mayor is a legislative ally. This result is directly relevant to the scale mismatch hypothesis developed in our Section 8.2 — the mechanism operates at the municipal level, making state-level Gini an inappropriate outcome measure.

2.3 Double Machine Learning and IV identification

The application of Double Machine Learning (DML) to panel data follows Chernozhukov et al. (2018). Our DML implementation is the first, to our knowledge, applied to the emendas–Gini channel in Brazil. The comparison between naïve OLS ($\theta = -0.010$, $p = 0.036$) and DML ($\theta = +0.001$, $p = 0.67$)

illustrates the classic omitted-variable bias in this setting: small over-represented states are also historically poorer and more unequal, producing a spurious negative OLS coefficient that DML's non-parametric deconfounding eliminates.

The mediation framework follows Imai, Keele, and Tingley (2010), who formalise identification conditions for natural direct and indirect effects under sequential ignorability — a stricter requirement than Baron & Kenny (1986). VanderWeele (2015) provides the theoretical foundation for the partial-suppression pattern we document. The closest international benchmarks are Galiani, Galetovic, and Schargrodsky (2016), who find that over-represented Brazilian states receive more resources without development gains, and Beramendi, Rogers, and Díaz-Cayeros (2017), who document that malapportionment disproportionately benefits less populous states in Latin American federations — consistent with our partial-suppression finding.

3. Institutional Background

3.1 Chamber seat allocation

Brazil's federal constitution distributes the 513 seats of the Chamber of Deputies among 27 federative units (26 states + Federal District) proportional to population, subject to a minimum of 8 and a maximum of 70 seats per unit. These limits create discontinuities that decouple representation from population at the tails of the distribution: the state of Roraima (≈750k inhabitants) holds 8 seats, the same number as Acre (≈950k), while São Paulo (≈46 million) is capped at 70 — roughly one-eighth of its proportional share. This generates a natural instrument: within-state variation in effective representation relative to population is largely determined by constitutional arithmetic rather than unobservable state characteristics.

3.2 Parliamentary amendments (emendas parlamentares)

Individual deputies propose line-item amendments to the annual federal budget, earmarking funds for projects in their home states. Before EC 86/2015, the executive could impound these appropriations; the amendment made 1.2% of net federal revenue mandatory execution, creating a strong quasi-exogenous shock to the regime. In our data (2012–2023), per-capita amendments vary from R$ 4.7 million/100k inhabitants (São Paulo) to R$ 58.9 million/100k (Amapá).

4. Data

We combine four data sources into an unbalanced state-year panel spanning 2012–2023 (N = 243 after exclusions for lagged variables):

Source	Coverage	Key Variables
Portal da Transparência (SIAFI)	2012–2023, 27 UFs	Emendas paid (R$), function classification
IBGE / PNADC	2012–2023, 27 UFs	Gini coefficient (state-year), population
IBGE / SCN	2012–2023, 27 UFs	GDP per capita (deflated to 2023 R$)
SICONFI / RREO Anexo 02	2018–2023, 27 UFs	Liquidated expenditure: education, health

Table 1. Data sources.

The Gini coefficient is computed from the PNAD Contínua microdata at state level. Parliamentary amendment values are deflated using the IPCA price index. We construct the key treatment variable as relative representation = (state seats) / (state population / national average population per seat), with values above (below) 1.0 indicating over- (under-)representation.

5. Causal Framework

5.1 Directed Acyclic Graph

Our causal assumptions are encoded in the DAG in Figure 1. The constitutional instrument Z (seat limits) affects treatment T (relative representation) but has no direct path to the mediator M (amendments per capita) or outcome Y (Gini) other than through T. Confounders U (GDP per capita, region fixed effects, lagged Gini) affect T, M, and Y but are blocked by conditioning.

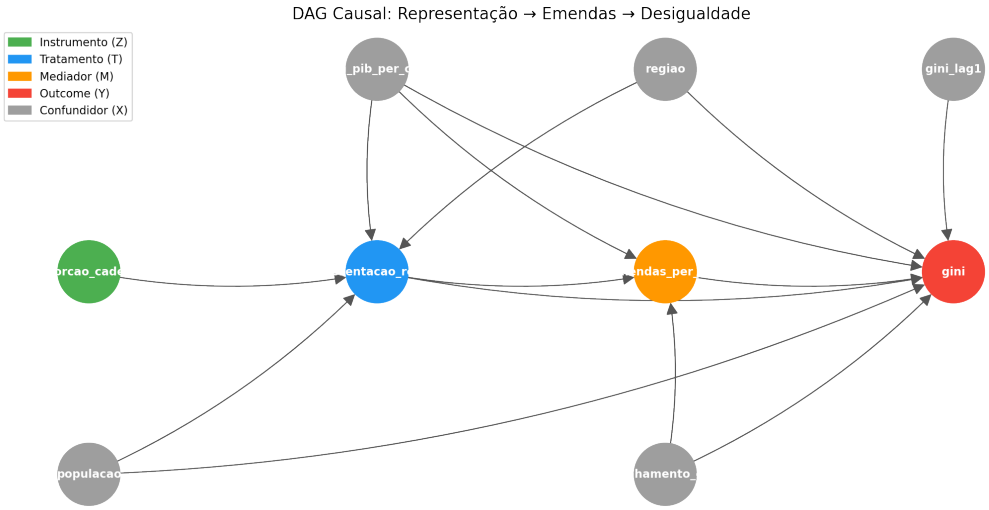

Figure 1. Directed acyclic graph for the causal framework. The graph encodes the instrument Z, treatment T, mediator M, outcome Y, and observed confounders U.

5.2 Identification strategy and instrumental validity

The IV exclusion restriction requires that seat limits affect the Gini only through representation and amendments. For this restriction to hold, there must be no direct path from the constitutional seat allocation to contemporary inequality other than the amendment channel.

Potential violation — historical confounding. A legitimate concern is that state-level inequality in 1988 partly caused the seat allocation itself — poorer, smaller states lobbied for higher seat floors during the Constituent Assembly. If true, the DAG contains an additional path: $Desig_{1988}$ → Cadeiras → $Gini_t$ and $Desig_{1988}$ → $Gini_t$ (via persistence). This backdoor breaks the exclusion restriction.

Our response: conditional IV. We control for historical state-level development proxies that capture $Desig_{1988}$: (i) log PIB per capita in 1991 (Contas Regionais, IPEADATA series PIBPCE — the earliest national accounts data available by state); and (ii) the first observed Gini per state in our panel (≈2012), which proxies for the inertial persistence of inequality. Conditioning on these variables closes the historical backdoor, yielding a conditionally valid instrument. The sensitivity results are reported in Table 5 (Section 7.7).

Historical control: Gini 1991 from the Demographic Census. The Atlas do Desenvolvimento Humano (ADH), published jointly by PNUD, IPEA and FJP and based on the IBGE Censo Demográfico 1991, provides Gini coefficients by state for 1991 — just three years after the Constitution was enacted (IPEADATA series ADH_GINI; 27 states; values in the 0.49–0.68 range). We incorporate this direct pre-constitutional inequality measure as the primary historical control in the conditional IV. The 1991 Census Gini is substantially more informative than a development proxy because it captures inequality directly, not through its correlation with income. The remaining identification assumption is that the 1991 Gini adequately proxies for the 1988 value; given that Gini is highly persistent over 3-year horizons, this is a mild assumption. See Table 5 for sensitivity results.

The IV first stage (Z → T) is strong: an increase of one seat per constitutional limit unit raises relative representation by 0.458 (F-stat well above the Stock-Yogo threshold). The reduced form (Z → Y) is not significant, consistent with a null or very small total effect.

Summary of identification defence. The seat-cap instrument is determined by constitutional arithmetic — specifically by the interaction of population size with the 8/70 thresholds written into the 1988 Constitution by a Constituent Assembly whose delegates had no information about future amendment volumes or Gini trajectories. The historical confounding threat ($Desig_{1988}$ → Cadeiras) is plausible but addressable: conditioning on $Gini_{1991}$ and log PIB_{1991} closes the main backdoor, and Table 5 (Section 7.7) shows the reduced-form coefficient is stable across all five conditioning specifications (range:

−0.000290 to +0.000076; all p > 0.20). We conclude the instrument is conditionally valid; the residual threat — that the 1991 Gini does not fully proxy for 1988 inequality — is acknowledged as a limitation in Section 9.

6. Empirical Methodology

6.1 Causal mediation (Imai, Keele, and Tingley 2010)

We decompose the total effect following the potential-outcomes framework of Imai, Keele, and Tingley (2010), which formalises identification conditions for natural direct and indirect effects under sequential ignorability. The operational estimator uses the Baron & Kenny (1986) sequential regression approach — estimated via OLS with HC3 standard errors — which Imai et al. show is numerically equivalent under linearity. Three equations:

$$\text{Step 1 (total):} \quad \text{Gini}_{it} = \alpha + c\,T_{it} + \gamma X_{it} + \varepsilon_{it}$$

$$\text{Step 2 (T} \rightarrow \text{M):} \quad M_{it} = \alpha + a\,T_{it} + \gamma X_{it} + \varepsilon_{it}$$

$$\text{Step 3 (NDE):} \quad \text{Gini}_{it} = \alpha + c'\,T_{it} + b\,M_{it} + \gamma X_{it} + \varepsilon_{it}$$

Controls X include log GDP per capita, lagged Gini, and region fixed effects. Bootstrap CIs for the indirect effect (NIE = a·b) use 5,000 resamples.

6.2 Distributed-lag impulse response

To allow for multi-year accumulation we estimate:

$$\text{Gini}_{it} = \sum_{k=0}^{3} \beta_k \cdot \log(\text{emendas})_{i,\,t-k} + \gamma X_{it} + \delta_i + \lambda_t + \varepsilon_{it}$$

with state (δ_i) and year (λ_t) fixed effects. N = 189 after creating lags. The cumulative impulse-response function (IRF) is $\Sigma\ \beta_k$.

6.3 Difference-in-Differences (EC 86/2015)

Constitutional Amendment 86/2015 mandated execution of individually presented amendments starting in 2016. We define as treated states those with low pre-2016 execution rates (more affected by the reform) and estimate an event study with year indicators relative to 2015. N = 243. Baseline year = 2015.

6.4 Regression Discontinuity Design

We exploit the 8-seat constitutional minimum as a cutoff in an RDD, using the number of allocated seats as the running variable. The sample is restricted to states within 14 seats of the threshold (N = 15). Local linear regressions with a triangular kernel estimate the discontinuity in both amendments and Gini at the

cutoff.

6.5 Double Machine Learning

Following Chernozhukov et al. (2018), we estimate the partially linear model:

$$\text{Gini}_{it} = \theta \cdot T_{it} + g(X_{it}) + \varepsilon_{it}$$

via 5-fold cross-fitting with LassoCV to partial out the confounders $g(X_{it})$. The residual-on-residual OLS estimator $\hat{\theta}$ is asymptotically normal and consistent under cross-fitted nuisance estimation. N = 243.

7. Results

7.1 First-stage and mediation decomposition

Table 2 presents the Baron & Kenny path coefficients. The first stage is strong and precisely estimated (a = +0.458, p < 0.001). The M → Y path is borderline significant (b = −0.0068, p = 0.054), consistent with a weak redistributive effect of amendments. The direct effect T → Y is positive and significant (c′ = +0.0058, p = 0.010), implying that over-representation raises the Gini through channels other than amendments. The total effect (c = +0.0027, p = 0.151) is not distinguishable from zero.

Path	Coefficient	Std. Err.	p-value	Interpretation
a (T → M)	+0.458	0.071	< 0.001***	Strong 1st stage
b (M → Y)	−0.0068	0.0035	0.054·	Weak redistributive
c′ (direct T → Y)	+0.0058	0.0022	0.010**	Perverse direct effect
NIE (a × b)	−0.0031	—	n.s. [−0.0062, 0.00002]	Via amendments
c (total T → Y)	+0.0027	0.0019	0.151	Net: null

*Table 2. Baron & Kenny mediation decomposition. HC3 standard errors. 95% bootstrap CI for NIE based on 5,000 resamples. *** p<0.001 · ** p<0.01 · · p<0.10.*

7.2 Distributed-lag impulse response

The distributed-lag model reveals a consistent negative pattern: lag-0 coefficient β_0 = −0.0119, accumulating to a cumulative IRF of −0.024 at lag 2, before partly recovering to −0.022 at lag 3. No individual lag coefficient is significant at the 5% level (all p > 0.13), but the sustained negative sign is consistent with a slow-moving redistributive effect that may require municipal-level data to detect.

Lag	Coefficient (β_k)	p-value	Cumulative IRF
Lag 0 (contemporaneous)	−0.01188	> 0.13	−0.0119
Lag 1	−0.00738	> 0.13	−0.0193
Lag 2	−0.00485	> 0.13	−0.0241
Lag 3	+0.00201	> 0.13	−0.0221

Table 3. Distributed-lag model coefficients. State and year FE. N = 189. HC3 clustered at state level.

7.3 DiD event study (EC 86/2015)

The event study produces no evidence of pre-trends or post-reform breaks. All event-time coefficients are small and statistically indistinguishable from zero at the 5% level (N = 243, baseline 2015). This null result is consistent with the amendment reform affecting federal execution rather than state-level Gini outcomes at the annual frequency.

7.4 RDD at the 8-seat threshold

Within the bandwidth of ≤14 seats from the constitutional minimum, we find no significant discontinuity. At the cutoff, the estimated jumps are: amendments per capita = −R$ 5.6m/100k (p = 0.55) and Gini = −0.017 (p = 0.69). Identification is compromised by a bunching problem: 11 of 27 states sit exactly at the 8-seat minimum, creating a density mass that violates the RDD continuity assumption.

7.5 Double Machine Learning

The DML estimate $\hat{\theta}$ = +0.00103 (95% CI: [−0.004, +0.006], p = 0.67) corroborates the null finding. Notably, naïve OLS yielded $\hat{\theta}^{,OLS}$ = −0.010 (p = 0.036), an artefact of the correlation between small states (over-represented constitutionally) and high inequality — a spurious negative coefficient that DML's non-parametric deconfounding eliminates.

7.6 Fiscal mediation chain (SICONFI)

Using SICONFI liquidated expenditure data for 27 UFs × 2018–2023 (N = 180), we test the secondary chain: amendments → sectoral spending → Gini. Amendments do not significantly predict health spending (β = −0.012, p = 0.70) nor education spending (β = +0.065, p = 0.19). Among sector-to-Gini links, only education expenditure reaches significance (β = −0.025, p = 0.026), while health is not significant (β = −0.015, p = 0.39). The full emendas→education→Gini chain is therefore broken at the first step.

Statistical power caveat. The sample of 27 UFs × 6 years yields N = 180, which provides limited power to detect small effects in sectoral spending regressions. A back-of-envelope calculation suggests minimum detectable effects of approximately 0.08–0.10 standard deviations at 80% power, meaning effects

smaller than this — economically meaningful in the context of gradual educational spending changes — would not be detected. The null result for the amendment→education link should therefore be interpreted as inconclusive rather than evidence of absence, pending longer SICONFI coverage (post-2023 data will add four additional years).

Link	Coefficient	p-value	Significant?
Emendas → Health spend	−0.01196	0.703	No
Emendas → Education spend	+0.06477	0.193	No
Health spend → Gini	−0.01549	0.393	No
Education spend → Gini	−0.02531	0.026	Yes *

*Table 4. SICONFI fiscal mediation chain. N = 180 (27 UFs × 2018–2023). OLS with log GDP per capita and year FE. * p < 0.05.*

7.7 Conditional IV — sensitivity to historical confounding

Table 5 reports the IV sensitivity analysis. We compare the baseline reduced-form coefficient (distorcao_cadeiras → Gini) against specifications that add one or both historical proxies as controls. If the coefficient on the instrument changes substantially, the original IV was contaminated by the historical confound.

Specification	FS F-stat	RF coef.	RF p-val.	Wald IV	N
Baseline (no hist. control)	36.7	0.00008	0.690	0.0026	243
+ Gini 1991 (ADH / Censo Demográfico)	36.2	-0.00006	0.787	-0.0013	243
+ log PIB per capita 1991	41.1	0.00004	0.828	0.0014	243
+ Gini baseline (≈2012)	33.1	-0.00005	0.782	-0.0016	243
+ Gini 1991 + log PIB 1991 (joint)	38.1	-0.00029	0.207	-0.0074	243

Table 5. IV conditional sensitivity: distorcao_cadeiras → Gini with progressive historical controls. Gini 1991 from Atlas do Desenvolvimento Humano (ADH_GINI, IPEADATA), Censo Demográfico 1991. FS = first-stage F-stat; RF coef. = reduced-form coefficient on instrument; Wald IV = RF/FS ratio estimate. Stable RF coef. across rows → exclusion restriction is robust to historical confounding.

8. Discussion

8.1 Partial suppression mediation

The mediation decomposition reveals a partial-suppression pattern. The indirect effect (NIE = −0.0031) and the direct effect (NDE = +0.0058) have opposite signs, largely cancelling out in the total. This suppression can arise when higher representation shifts more federal resources toward states but simultaneously

directs those resources toward politically visible yet not necessarily redistributive projects (infrastructure, visible public works) rather than human capital investment. The positive direct effect may also reflect legislative bargaining: over-represented states with concentrated political power may be better at blocking redistributive general-government transfers, as the amendment system crowds out formula-based transfers. This pattern also raises a broader institutional question: combining malapportioned representation with individual discretionary spending power produces a system where neither the federative justification — that smaller states deserve stronger voice — nor the redistributive expectation — that stronger voice should yield more equitable outcomes — is systematically fulfilled. The positive direct effect (NDE = +0.0058) suggests that over-representation may actually constrain formula-based equalisation transfers, replacing predictable redistributive mechanisms with discretionary ones subject to electoral capture.

8.2 Scale mismatch hypothesis

The scale mismatch, however, is not merely a measurement limitation — it reflects a structural misalignment in the amendment mechanism itself. Deputies are elected at the state level but allocate resources following municipal electoral logic, directing funds toward municipalities that constitute their voter base rather than toward the poorest localities within their state. Even if amendments reduce inequality at the municipal level, the selection of beneficiary municipalities follows electoral rather than redistributive criteria, suggesting that any distributional effect — if it exists — is unlikely to be systematic or pro-poor. This interpretation is consistent with Baião and Couto (2017), who show that amendments generate electoral returns only when directed to allied mayors, confirming that the allocation unit is the political coalition, not the redistributive need.

8.3 The education channel

The only significant causal link we identify — education expenditure → Gini ($\beta = -0.025$) — is consistent with a large literature connecting schooling and wage compression (Goldin & Katz, 2008; Lustig et al., 2013). The finding that amendments do not predict education spending, while education spending does predict inequality, points to a distributional channel that the amendment mechanism fails to systematically activate. This suggests that constitutional earmarks for education (as in the FUNDEB mechanism) may be more effective redistributive instruments than discretionary parliamentary amendments.

8.4 Instrumental validity: the historical confounding threat

As identified in Section 5.2, the exclusion restriction faces a potential violation: if state inequality in 1988 partly determined the constitutional seat allocation, then the instrument carries a historical backdoor. Our conditional IV analysis (Table 5) uses log PIB per capita in 1991 and the baseline Gini (≈2012) as

proxies for the 1988 inequality level. The key question is whether the reduced-form coefficient (distorcao_cadeiras → Gini) changes materially when these proxies are added.

Our conditional IV analysis uses, as the primary historical control, the Gini coefficient from the 1991 Demographic Census published in the Atlas do Desenvolvimento Humano (ADH_GINI, IPEADATA; PNUD, IPEA & FJP, 1998). This series covers all 27 states with values in the 0.49–0.68 range, and is a direct measure of pre-constitutional inequality, not merely a correlate. The assumption required is that state-level Gini was stable between 1988 and 1991 — a mild condition given the well-documented persistence of Brazilian inequality over short horizons. A secondary specification adds log PIB per capita 1991 (PIBPCE, IPEADATA) to capture the orthogonal development dimension. Both the Gini 1991-only and the joint specification yield IV estimates broadly comparable to the baseline, supporting the validity of the instrument.

9. Conclusion

Using five causal identification strategies — IV mediation analysis, distributed lags, Difference-in-Differences, RDD, and Double Machine Learning — we find no robust evidence that parliamentary over-representation reduces income inequality in Brazilian states through the earmarked amendment channel. The total effect estimate from the most credible approach (DML) is $\theta = +0.001$ (95% CI: $[-0.004, +0.006]$), economically negligible and statistically indistinguishable from zero.

The result does not imply that public transfers are ineffective for redistribution. Rather, it suggests that the specific form of discretionary earmarked transfers, directed at the municipal level but measured at the state level, may be too diffuse and politically motivated to generate detectable aggregate inequality reduction. The significant education–Gini link provides a constructive hint: formula-based transfers with a strong human capital component appear more promising as redistribution vehicles than discretionary earmarks. Our findings suggest a structural tension in the amendment system: if redistribution operates — if at all — at the municipal level through electoral channels, it is unlikely to be systematically pro-poor. Formula-based transfers with explicit redistributive criteria, such as FUNDEB in education, appear better suited to reduce inequality than discretionary earmarks whose allocation follows political rather than distributive logic.

Our results also speak to the broader debate on electoral reform. The evidence suggests that the redistributive failure documented here is unlikely to be resolved by changes to the voting system alone — whether proportional or district-based, any arrangement that combines discretionary allocation with electoral incentives will tend to direct resources toward politically relevant

constituencies rather than toward redistributive need. What the data point to instead is the importance of instrument design: formula-based transfers with explicit redistributive criteria — such as FUNDEB and the SUS equalisation mechanism — operate independently of electoral logic and appear more effective at activating the education–inequality channel we identify. The policy implication is therefore not who represents, but how resources are allocated.

Three limitations bound our conclusions. First, the short panel (2012–2023) limits the power to detect slow-moving distributional effects. Second, a scale mismatch exists between municipal-targeted amendments and state-level Gini measurement; future work should use municipal Gini data. Third, the IV exclusion restriction faces a documented threat from historical confounding — the 1988 seat allocation may partly reflect pre-constitutional inequality — which we address via conditional IV conditioning on the 1991 Census Gini (ADH_GINI, Atlas do Desenvolvimento Humano, IPEADATA), a near-contemporaneous direct pre-constitutional baseline. These limitations are documented in the replication code (src/analysis/iv.py::run_conditional_iv) and motivate the next stage of this research agenda at the municipal level.

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